

Producer Theory

Lecture 15: Elasticity — Demand Recap & Supply Elasticity

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Recap: Previous Lectures

Lecture 1: Economics, Scarcity, Efficiency, Macro vs. Micro

Lecture 2: Three economic problems (WHAT, HOW, FOR WHOM) and economic systems (Market, Planned, Mixed)

Today's Focus:

 How do people make economic decisions?

 What is rationality?

 **Opportunity cost** — one of economics' most powerful concepts!

Economic Rationality

What Does “Rational” Mean?

ECONOMIC RATIONALITY

People are **rational decision-makers** who seek to maximize their well-being (utility) by using cost-benefit analysis and responding predictably to incentives.

Key characteristics:

- ✓ People have **clear preferences**
- ✓ They seek to **maximize their well-being** (utility)
- ✓ They respond **predictably** to incentives
- ✓ They use **cost-benefit analysis**

 **Important:** Rational $\neq$ Perfect or emotionless! People can be rational within their information and cognitive limits (bounded rationality)

Cost-Benefit Principle

THE FOUNDATION OF RATIONAL CHOICE

An action should be undertaken **if and only if** the benefits exceed the costs.

Decision Rule:

$$\text{Decision Rule: } \begin{cases} \text{Do it} & \text{if } \textit{Benefits} \geq \textit{Costs} \\ \text{Don't do it} & \text{if } \textit{Benefits} < \textit{Costs} \end{cases}$$

In practice:

- Benefits = What you gain
- Costs = What you give up
- Compare net benefit of alternatives

Tourism Example: Cost-Benefit

Should a tourist visit the Belém Tower in Lisbon?

Benefits

 Photos & memories

 Educational value

 Historical experience

 Satisfaction

Estimated value: €20

Costs

 Entry ticket: €6

 Travel time: 30 min

 Waiting time: 20 min

 Value of time: €15/hr

Time cost: €12.50

Total cost: €18.50

 **Decision:** Benefits (€20) > Costs (€18.50) → **Visit!**

But if waiting time increases to 60 min, costs rise to €28.50 → **Don't visit!**

Marginal Analysis

Decisions at the Margin

MARGINAL ANALYSIS

Decisions are made **at the margin** — evaluating the additional benefit versus the additional cost of one more unit.

Key terms:

 **Marginal Benefit (MB)**: Benefit from one additional unit

 **Marginal Cost (MC)**: Cost of one additional unit

Optimal Decision Rule:

Continue activity while $MB \geq MC$

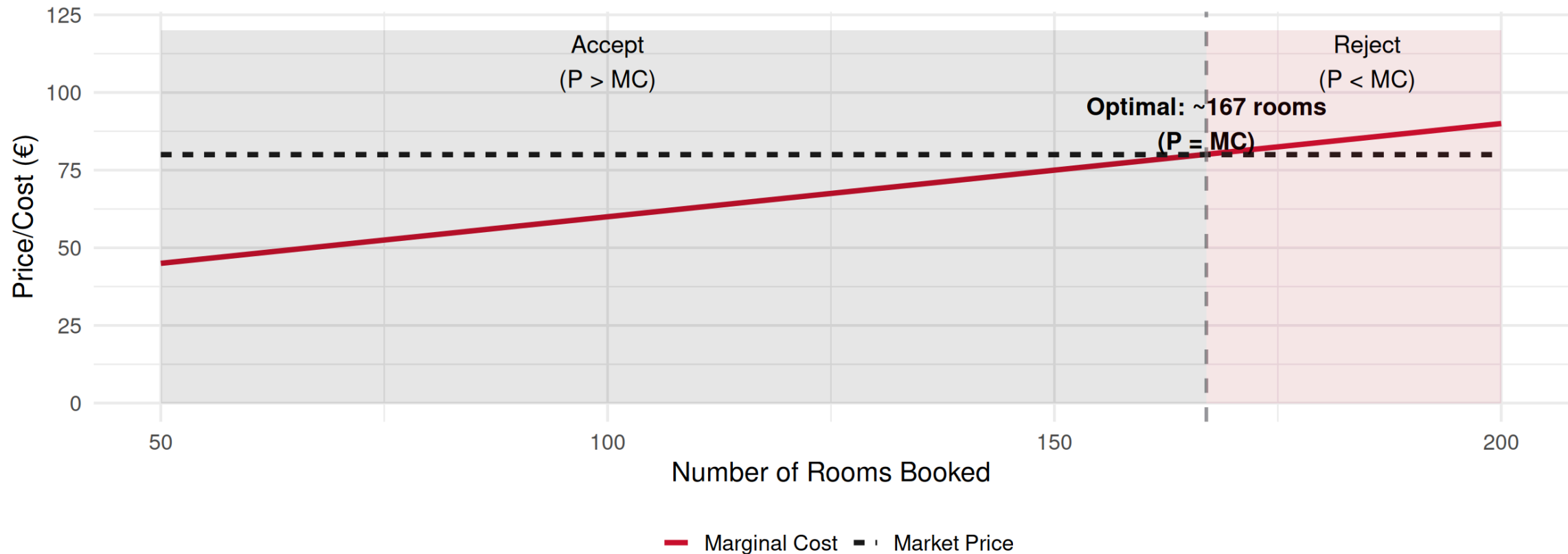
Stop when $MB < MC$

Marginal Analysis: Hotel Rooms

Should a hotel accept one more booking?

Hotel Room Pricing: Marginal Analysis

Accept bookings while Price > Marginal Cost



Hypothetical illustration of marginal decision-making

Opportunity Cost

The Most Important Concept!

OPPORTUNITY COST

The value of the **best alternative foregone** when making a choice. It's what you give up, not what you pay.

Key Points

-  It's what you **give up**, not what you pay
-  It's the **next-best** alternative (not all alternatives)
-  It includes both **explicit** (money) and **implicit** costs (time, foregone opportunities)

Decision Rule with Opportunity Cost

When deciding: You should choose option A over B if:

$$\begin{array}{c}
 \text{Gross Benefit of A} \\
 \underbrace{B_A}
 \end{array}
 \geq
 \begin{array}{c}
 \text{Monetary Cost} \\
 \underbrace{C_A}
 \end{array}
 +
 \underbrace{(B_B - C_B)}_{\text{Surplus of best alternative}}$$

Opportunity Cost of A

💡 In simpler terms: Choose A if its benefits exceed both its direct costs AND what you're giving up from the best alternative!

Opportunity Cost Examples

Simple Example: You have €100 and 4 hours free on Saturday afternoon.

Options (value, cost, time):

- A. Visit museum (€15, €10, 3h)
- B. Beach trip (€20, €0, 4h)
- C. Movie (€22, €12, 2h)
- D. Study at home (€10, €0, 4h)

If you choose A (museum):

 **Explicit cost:** €10

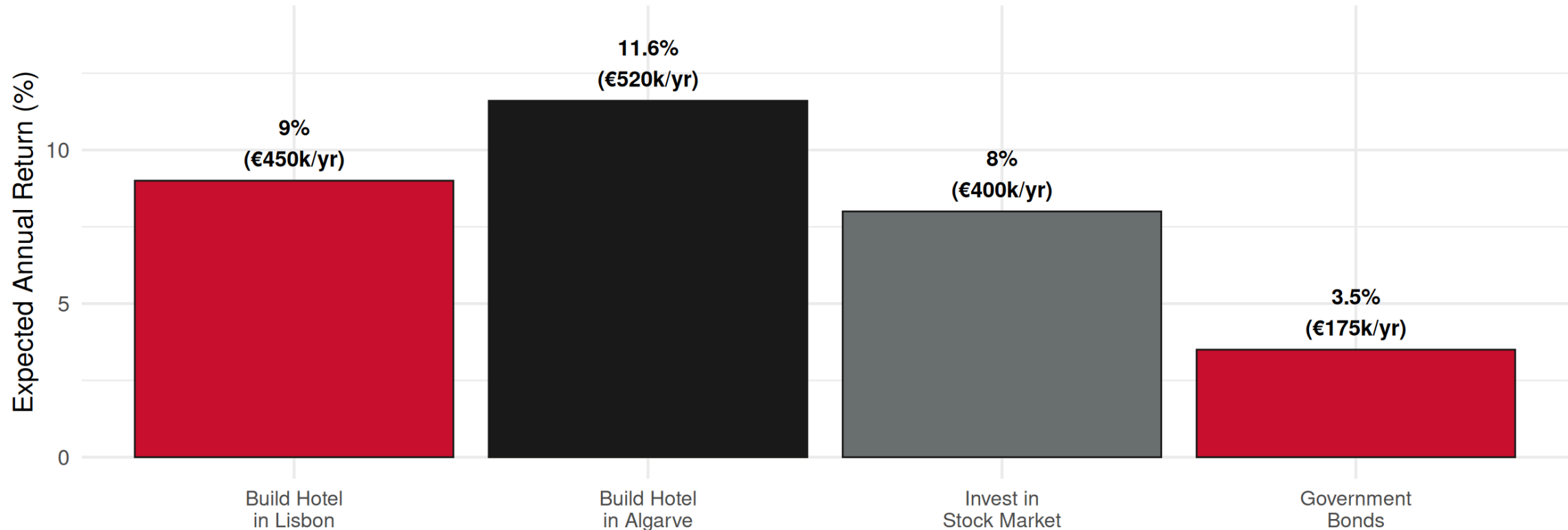
 **Opportunity cost:** Value of best alternative foregone

- Can't go to beach (value: €20)
- **Total opportunity cost:** €10 + €20 = **€30**

Opportunity Cost in Tourism Investment

Investment Alternatives: Expected Returns

Opportunity cost of choosing Lisbon hotel = foregone €520k from Algarve (best alternative)



Hypothetical illustration of investment alternatives

Sunk Costs: Ignore Them!

SUNK COST

A cost that has already been incurred and **cannot be recovered**, regardless of what decision you make now. Sunk costs should be **ignored** in decision-making!

Examples of sunk costs:

-  Non-refundable deposits
-  Past advertising expenses
-  Money already spent on failed projects
-  Time already invested (can't get it back)

 **Common mistake:** “I’ve already invested so much, I must continue!”

 **Correct thinking:** “What are my future costs and benefits from here?”

Sunk Cost Example: Tourism Investment

Scenario: Hotel construction halfway done

Spent so far (SUNK):

- €2M on land
- €3M on construction
- **Total:** €5M

 Cannot be recovered!

 **Correct analysis:** Ignore the €5M sunk cost!

Compare: €4M future cost vs. future benefits

If NPV of future benefits > €4M → **Finish it!**

If NPV of future benefits < €4M → **Abandon!** (Don't throw good money after bad)

To complete (FUTURE):

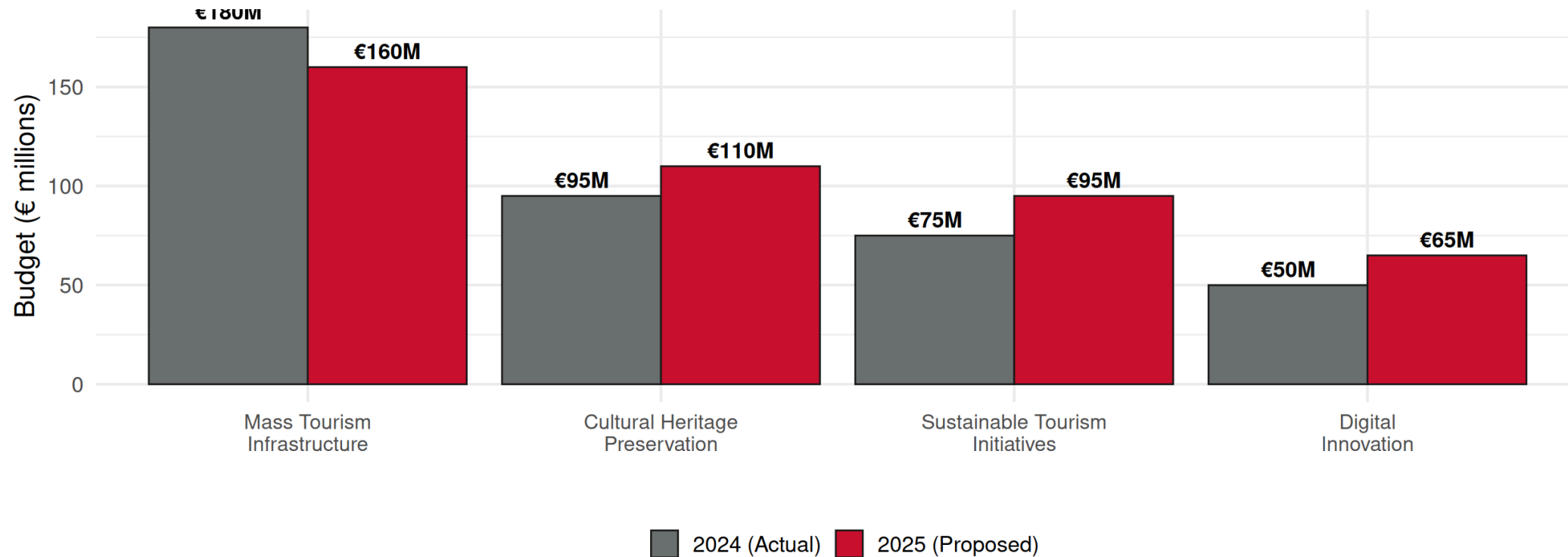
- €4M more needed
- Expected profit: €500k/year

 Should you finish?

Trade-offs in Tourism Policy

Tourism Investment Allocation: Wakanda Government

Trade-offs: More cultural preservation = Less mass tourism infrastructure (hypothetical)



👉 **Opportunity cost of increasing cultural preservation by €15M = €20M less for mass tourism infrastructure**

Making Better Decisions

Checklist for Rational Economic Decisions:

- 1 Identify **all alternatives**
 - 2 Estimate **benefits** of each alternative
 - 3 Estimate **costs** (including opportunity costs!)
 - 4 **Ignore sunk costs** (they're irrelevant)
 - 5 Consider **marginal** changes (not just totals)
 - 6 Choose alternative with **highest net benefit**
-  **Remember:** Good decisions maximize net benefit (Benefits - Opportunity Costs)

Common Decision Mistakes

Mistake 1: Ignoring opportunity costs

- Thinking only about money paid
- Not considering value of alternatives foregone

Mistake 2: Counting sunk costs

- “I’ve already invested so much”
- Looking backward instead of forward

Mistake 3: Ignoring marginal analysis

- Looking at totals instead of additional units
- Not considering “one more” decisions

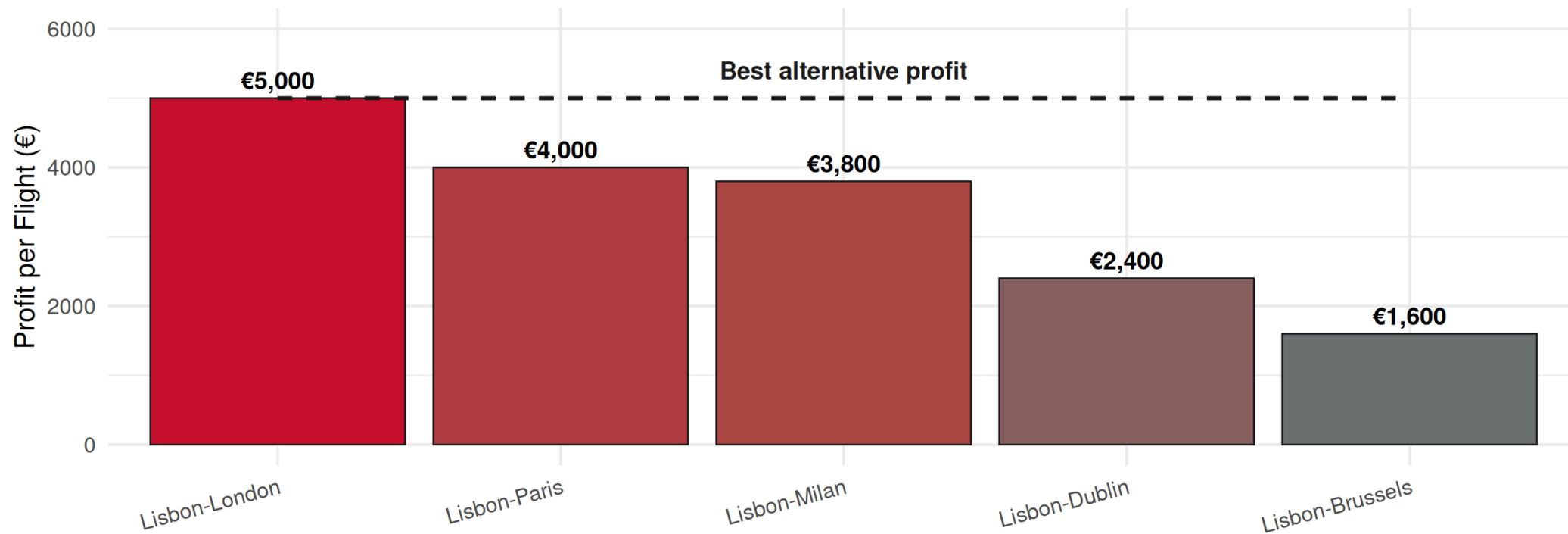
Mistake 4: Not comparing net benefits

- Choosing highest benefit option
- But ignoring that it also has highest cost!

Real Business Case: Ryanair Strategy

Ryanair Route Profitability Analysis

Opportunity cost: Each flight to Brussels = foregone €3,400 from London route (hypothetical)



👉 **Decision:** Cancel Brussels route, add London frequency → Better use of aircraft (opportunity cost logic)

Key Concepts Summary

Concept	Definition
Rationality	People maximize well-being via cost-benefit analysis
Cost-Benefit Principle	Do it if benefits $\geq$ costs
Marginal Analysis	Decisions at the margin (additional units)
Opportunity Cost	Value of best alternative foregone
Sunk Costs	Ignore them! (already spent, irrelevant)
Trade-offs	Choosing one thing means giving up another

Exercises



Practice Time!

Real-world applications of opportunity cost.

Exercise 1: Multiple Choice

A hotel owner can use her restaurant space for:

- Option A: Fine dining (€80k annual profit)
- Option B: Buffet restaurant (€60k annual profit)
- Option C: Rent to external operator (€50k annual rent)

She chooses Option A. What is the opportunity cost of this decision?

- A. €80,000
- B. €60,000
- C. €50,000
- D. €110,000

Answer: B (€60,000) - Opportunity cost = value of best alternative foregone. The next-best alternative to fine dining (€80k) is buffet (€60k), NOT the sum of all alternatives.

Exercise 2: Multiple Choice

You bought a non-refundable €200 ticket to a concert. On the day of the concert, you feel ill. A friend offers you €50 for the ticket (transferable). What is the opportunity cost of **ATTENDING** the concert?

- A. €200
- B. €150
- C. €50 + discomfort from illness
- D. €250

Answer: C (€50 + discomfort) - The €200 is SUNK (can't recover even if you don't go). If you attend: you give up €50 (could sell) + endure discomfort. Opportunity cost = what you give up by attending.

Exercise 3: Open Question

Scenario: A tour operator in Porto must decide between two summer season strategies:

Strategy A: Focus on luxury tours

- Expected revenue: €500,000
- Variable costs: €280,000
- Fixed costs already committed: €80,000

Strategy B: Focus on budget tours

- Expected revenue: €420,000
- Variable costs: €200,000
- Same fixed costs: €80,000

Questions:

- a. Calculate profit for each strategy
- b. What is the opportunity cost of choosing Strategy A?
- c. Which strategy should be chosen and why?
- d. Are fixed costs relevant to the decision? Explain.

Exercise 3: Solution (Part a & b)

a) Profit calculation:

Strategy A (Luxury):

$$\text{Profit}_A = €500,000 - €280,000 - €80,000 = €140,000$$

Strategy B (Budget):

$$\text{Profit}_B = €420,000 - €200,000 - €80,000 = €140,000$$

Both strategies have equal profit!

b) Opportunity cost of Strategy A:

Opportunity cost = Direct Cost + Profit from best alternative foregone = **€140,000 + €360,000 = €500,000** (Strategy B profit)

Exercise 3: Solution (Part c & d)

c) Which strategy to choose?

From pure profit perspective: **Indifferent** (both €140,000)

But consider other factors: Risk, reputation, market trends, resource requirements

👉 **Recommendation: Strategy B (budget)** because:

- Lower variable costs (€200k vs €280k) = **less risk**
- Higher revenue-to-variable-cost ratio: **2.1 vs. 1.79**
- More resilient to demand fluctuations

d) Are fixed costs relevant?

✗ **NO!** Fixed costs (€80,000) are **SUNK** — already committed regardless of strategy choice.

✓ Only **variable costs** matter for this decision (they differ between strategies).

This is a key example of ignoring sunk costs in decision-making!

Next Lecture

Lecture 4 (February 13, 2026):

- **Production Possibilities Frontier (PPF)**
- Efficiency, Trade-offs, and Economic Growth
- Comparative Advantage and Specialization

 **Preparation:** Review today's opportunity cost concept — it's the foundation for understanding PPF!

Thank You! 🖐️

Questions?

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Next class: Thursday, February 13, 2026